Functional Requirements Document

Project 2 · Loan Origination System (LOS) Transformation

Purpose. Translate the business goals into functional behavior and testable requirements for the proposed digital loan application process. Items not specified in the project summary are identified for confirmation.

1. Solution overview

The proposed LOS workflow should coordinate application intake, completeness checks, eligibility assessment, manual review where required, decision recording and applicant communication. It should reduce unnecessary handoffs, make exceptions visible and provide consistent data for turnaround-time and rejection analysis. This FRD defines expected behavior without assuming a particular LOS product or architecture.

Implementation dependency Confirm the target LOS platform, existing system interfaces, decision-engine capabilities and which steps are automated versus performed by authorized staff.

2. User roles

Role	Expected access / actions
Applicant / channel user	Submit application information and supporting documents; receive status or information requests through approved channels.
Loan operations user	Review completeness, manage missing-information requests, update operational status and route cases.
Credit analyst / underwriter	Review eligible cases, assess evidence, record recommendations and refer exceptions.
Credit decision authority	Approve, decline or return applications within assigned authority and policy.
Risk / compliance reviewer	Review permitted cases, controls, exceptions and audit evidence.
System administrator	Maintain approved reference data, routing configuration and access settings under change control.
Reporting analyst / auditor	Access authorized aggregate metrics and audit records.

3. Functional requirements

ID	Requirement	Acceptance criteria
FR-01	Register a submitted loan application.	A unique application ID is generated; submission timestamp and channel are recorded; duplicate handling follows an agreed rule.
FR-02	Validate mandatory application fields.	Missing or invalid required fields are identified; the user receives actionable feedback or the case is routed to a defined incomplete state.
FR-03	Capture and track supporting documents.	Configured document types show required/received/accepted status; missing or unreadable items can be flagged for follow-up.
FR-04	Maintain application status through the journey.	Statuses and permitted transitions are defined; each status change records actor/system and timestamp.
FR-05	Perform approved eligibility and business-rule checks.	The system evaluates only approved, versioned rules; each outcome is recorded with rule/version reference where supported.
FR-06	Route applications to the appropriate next step.	Routing follows agreed product, risk, completeness and exception criteria; no case is left without a valid queue or owner.
FR-07	Support manual underwriting and exception review.	Authorized reviewers can record findings, request additional information, refer a case or submit a decision within their permissions.

FR-08	Record the final decision.	Decision value, decision timestamp, decision authority and required rationale are captured; unauthorized users cannot finalize a decision.
FR-09	Communicate missing-information requests and decisions.	Approved templates are used; recipient, event and delivery status are logged where available; sensitive details follow communication policy.
FR-10	Track processing time and SLA status.	The agreed start/end events are timestamped; elapsed time and overdue status can be calculated consistently.
FR-11	Capture rejection reasons using controlled categories.	Decline reasons are selected from an approved taxonomy; mandatory rationale and any required disclosures are enforced by policy.
FR-12	Provide an operational work queue.	Authorized users can filter cases by status, age, owner, product, pending action and SLA status, subject to access permissions.
FR-13	Maintain audit history.	Key field changes, status transitions, rule outcomes, approvals and decisions are traceable with timestamps and actors.
FR-14	Support exception and failure handling.	Integration, document-processing or rule-evaluation failures are visible, logged and routed to an agreed recovery process.
FR-15	Provide data for operational reporting.	Authorized users can obtain the fields needed for turnaround time, rejection trends, bottleneck and workflow-volume analysis.
FR-16	Track rule changes through controlled governance.	Changes to decision rules or reference values require authorized approval, version tracking and an effective date.

4. Proposed status model

Status	Meaning / exit condition
Submitted	Application received and assigned a unique reference.
Completeness Check	Required fields and documents are being validated.
Information Required	Applicant or internal party must provide missing information.
Ready for Assessment	Minimum required information is present and validation has passed.
Assessment in Progress	Automated checks and/or analyst assessment are underway.
Manual Review	Case requires authorized human review or exception handling.
Decision Pending	Required assessment is complete and final authorized decision is awaited.
Approved	An authorized approval has been recorded.
Declined	An authorized decline and required reason have been recorded.
Withdrawn / Cancelled	Application is closed without a lending decision under an approved closure rule.

This is a proposed model for review. The final status names, transitions and permitted reopen/rework paths must be agreed with operations and credit stakeholders.

5. Business rules and controls

ID	Rule
RULE-01	Only approved and effective-dated business rules may be used to drive eligibility or decision routing.
RULE-02	A final decision must be made by an authorized role or approved automated mechanism within defined policy limits.
RULE-03	Applications with required information missing must follow the defined incomplete-application path.
RULE-04	Manual overrides, referrals and exceptions must capture the required reason and approver.
RULE-05	Rejection categories must be consistent, reportable and governed by the designated business owner.
RULE-06	The turnaround-time calculation must use agreed start/end events and treatment of paused or waiting time.
RULE-07	Access to applicant data and decision information must be role-based and follow privacy/security policy.

RULE-08	A technical error must not silently produce an approval or decline; affected cases must be safely held or routed for recovery.

6. Data requirements

Data group	Illustrative fields	Validation / notes
Application	Application ID, submission date/time, channel, product type, application status	Confirm unique key, source of truth and duplicate policy.
Applicant	Applicant reference, required identity/contact attributes	Use only fields required by the approved process; apply privacy controls.
Documents	Document type, received date, validation status, exception reason	Confirm allowed formats, size limits and document retention rules.
Assessment	Check name, outcome, rule version, timestamp, referral flag	Use approved rules and keep decision logic traceable.
Decision	Decision, decision-maker, timestamp, rationale, reason code	Enforce decision authority and required reason/disclosure rules.
Process metrics	Stage timestamps, queue/owner, SLA target, pause reason	Define consistent events and time calculations.
Audit event	Actor, action, record reference, timestamp, before/after values where appropriate	Retention and access governed by policy.

7. Rejection-trend analysis

The project summary specifies SQL queries and Excel pivot tables for rejection analysis. The analysis should use agreed definitions and validated source data.

- Confirm the analysis population, time period, loan products and application status inclusion rules.
- Standardize rejection reason codes and identify null, generic, duplicate or inconsistent values.
- Use SQL to extract and validate counts, denominators and relevant segments; document joins and filters.
- Use Excel pivot tables to compare rejection counts and rates by reason, product, channel, period or other approved dimensions.
- Separate process-related rejections (for example, incomplete information) from credit-policy outcomes where the data supports that distinction.
- Review findings with credit and risk owners before proposing any rule change; correlation alone should not be treated as proof that a rule is incorrect.

8. Integrations and interfaces

Interface area	Expected interaction	Open question
Application intake channel	Receive application data and return a reference or validation feedback.	Which web, branch, broker or partner channels are in scope?
Document management	Store, retrieve and track supporting documents.	What document repository and scanning/validation tools exist?
Credit bureau / external checks	Request approved checks and record responses.	Which providers, consent rules, timeouts and retry policies apply?
Decision engine / rules service	Evaluate approved rules and return traceable outcomes.	Is a decision engine present, and who owns rule deployment?
Notification service	Send requests for information and decision communications.	Which channels and approved templates are required?
Data / reporting layer	Provide trusted data for operational metrics and rejection analysis.	Which database, access controls and refresh cadence are available?
JIRA	Track requirements, change requests, defects and delivery traceability.	Confirm the team's issue types and traceability approach.

9. Non-functional requirements

Area	Requirement
Security	Role-based access, least privilege, secure authentication and controlled access to applicant and decision data.
Privacy	Collect and retain only approved data; apply access, retention and deletion policies.
Auditability	Maintain traceable records for decisions, approvals, rule versions, status changes and exceptions.
Availability / recovery	Define service availability, backup, recovery and business-continuity targets during solution design.
Performance	Agree response-time and throughput targets for submission, validation and assessment; do not assume unprovided numeric thresholds.
Usability	Make current status, pending action, ownership and next steps clear to the relevant user.
Data quality	Validate mandatory fields, reference data, timestamps and reason codes before reporting.
Maintainability	Support controlled updates to routing rules, reason categories and notification templates.

10. Minimum test scenarios

Test ID	Scenario	Expected result
TC-01	Submit an application with valid mandatory data.	Unique application ID is created and submission time is recorded.
TC-02	Submit an application with a required field missing.	Validation identifies the issue and routes the case to the agreed incomplete path.
TC-03	Required supporting document is missing.	Document status is visible and the defined information-request process is triggered.
TC-04	Application passes completeness checks.	Case can move to Ready for Assessment according to the status model.
TC-05	Rule check returns an exception or cannot complete.	Case is routed to a safe exception path; no silent final decision occurs.
TC-06	Case requires manual underwriting.	Authorized reviewer can record findings and route or decide within assigned permissions.
TC-07	Unauthorized user attempts to finalize a decision.	Action is denied and handled according to the security/audit model.
TC-08	Authorized decline is recorded.	Decision, reason category, authority and timestamp are stored as required.
TC-09	Application exceeds configured SLA.	Case is identified as overdue and appears in the appropriate queue/report.
TC-10	External service or notification fails.	Failure is logged and retry/escalation or manual recovery is available.
TC-11	Compare rejection report with source records.	Counts, rates, filters and categories reconcile to validated source data.
TC-12	Review an application's audit trail.	Authorized reviewer can trace relevant status changes, rule outcomes and decisions.

11. Open questions for discovery

- Which loan products and application channels are included in the transformation?
- What is the current LOS and which systems provide applicant, document, bureau and decision data?
- What are the four identified bottlenecks, and what evidence supports each one?
- How is loan decision turnaround time defined, and what baseline/cohort supports the reported 20% improvement?
- Which rejection reason taxonomy and credit-rule governance process are authoritative?
- Which stages may be automated, and which require human review or approval?

- What SLAs, escalation paths, retention periods and non-functional targets must be met?

12. Functional sign-off

The FRD is ready for sign-off when operations, credit, risk, development and QA agree that the requirements are feasible, testable and aligned with approved policy.

Role	Name	Decision / date
Business analyst / requirements owner	To be completed	To be completed
Loan operations owner	To be completed	To be completed
Credit / risk representative	To be completed	To be completed
Development lead	To be completed	To be completed
QA lead	To be completed	To be completed

Tools referenced

BPMN 2.0 · SQL · Excel Pivot Tables · Gap Analysis · Process Mapping · JIRA